

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 8 - Formative Assessment 2

Formative Assessment (FA-2)

Class: Class 8

Assessment Type: Formative Assessment (FA-2)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Data Structures - Lists, Tuples, Dictionaries

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: Written Concept Paper (20 Marks)

Section I: MCQ (5 x 1 = 5 marks)

- a. Which of the following creates an empty list in Python?
 - b. `my_list = ()`
 - c. `my_list = {}`
 - d. `my_list = []`
 - e. `my_list = dict()`
- b. What is the time complexity of accessing an element in a Python list by index?
 - c. $O(n)$
 - d. $O(1)$
 - e. $O(\log n)$
 - f. $O(n^2)$
- c. Which of the following is **immutable** in Python?
 - d. List
 - e. Dictionary
 - f. Tuple
 - g. Set
- d. What will be the output of:

```
d = {"a": 1, "b": 2}
d["c"] = 3
print(len(d))
```

 - a. 2
 - b. 3
 - c. Error
 - d. None
- e. Which method adds an element to the end of a list?
 - f. `add()`
 - g. `insert()`

h. `append()`

i. `extend()`

Section II: Short Answer (5 x 2 = 10 marks)

a. Differentiate between a **list** and a **tuple**. Give two use cases of each.

b. Write the output of:

```
my_list = [10, 20, 30, 40, 50]
print(my_list[1:4])
print(my_list[::-1])
```

c. Explain the difference between `append()` and `extend()` with examples.

d. Write a Python code snippet to iterate over all key-value pairs in a dictionary using the `items()` method.

e. What is a **nested dictionary**? Give one real-life example where it is useful.

Section III: Long Answer (5 x 1 = 5 marks)

a. Write a Python program that manages a student marks database using a dictionary of lists. The program should:

- Store marks for at least 5 subjects for 3 students
- Calculate the average mark per student
- Find the highest and lowest average
- Display the results in a formatted table

Part B: Hands-on Practical Lab Task (5 marks)

Write a Python program that:

- Creates a list of 10 integers entered by the user
- Uses list methods to: sort, reverse, find the maximum, find the minimum
- Computes the sum and average without using built-in `sum()` or `len()`
- Creates a tuple from the sorted list and displays both

Part C: Notebook Maintenance (5 marks)

- a. Neatness and organisation of notebook (2 marks)
- b. Completion of all assigned classwork and homework (2 marks)
- c. Proper headings, dates, and indexing (1 mark)